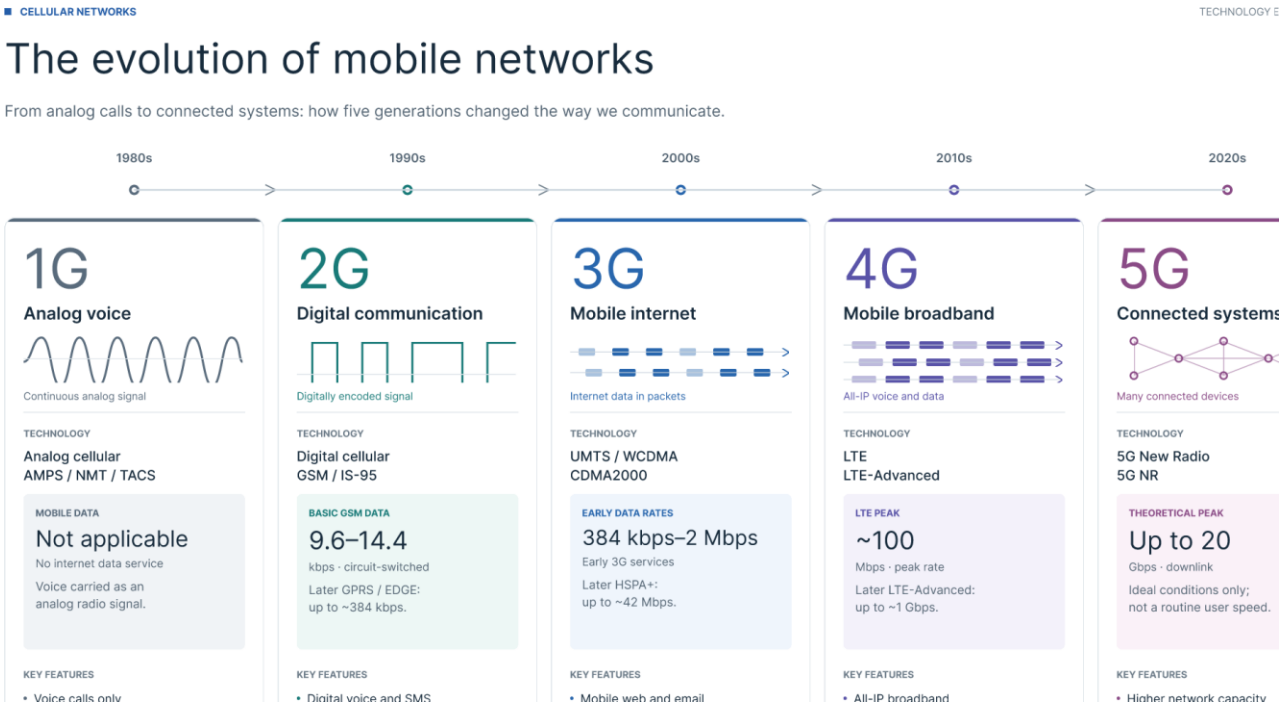
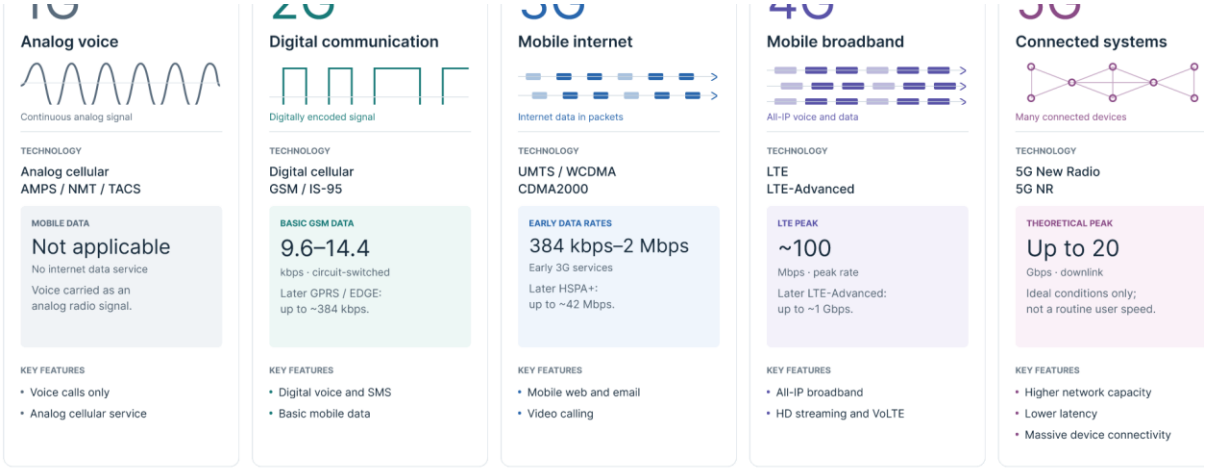
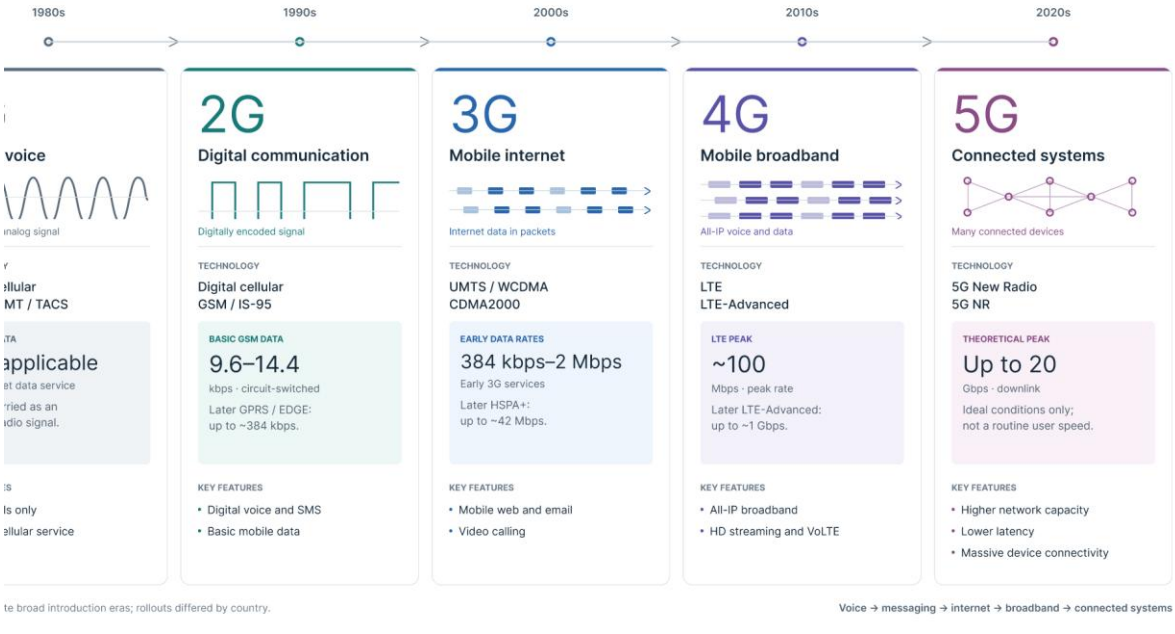
EXPERIMENT 12

Use Figma to create an interactive diagram illustrating the evolution from 1G to 5G




Analog voice
Continuous analog signal
TECHNOLOGY
Analog cellular
AMPS / NMT / TACS
MOBILE DATA
Not applicable
No internet data service
Voice carried as an analog radio signal.
KEY FEATURES
• Voice calls only
• Analog cellular service
Digital communication
Digitally encoded signal
TECHNOLOGY
Digital cellular
GSM / IS-95
BASIC GSM DATA
9.6–14.4
kbps · circuit-switched
Later GPRS / EDGE: up to ~384 kbps.
KEY FEATURES
• Digital voice and SMS
• Basic mobile data
Mobile internet
Internet data in packets
TECHNOLOGY
UMTS / WCDMA
CDMA2000
EARLY DATA RATES
384 kbps–2 Mbps
Early 3G services
Later HSPA+: up to ~42 Mbps.
KEY FEATURES
• Mobile web and email
• Video calling
Mobile broadband
All-IP voice and data
TECHNOLOGY
LTE
LTE-Advanced
LTE PEAK
~100
Mbps · peak rate
Later LTE-Advanced: up to ~1 Gbps.
KEY FEATURES
• All-IP broadband
• HD streaming and VoLTE
Connected systems
Many connected devices
TECHNOLOGY
5G New Radio
5G NR
THEORETICAL PEAK
Up to 20
Gbps · downlink
Ideal conditions only; not a routine user speed.
KEY FEATURES
• Higher network capacity
• Lower latency
• Massive device connectivity
Decades indicate broad introduction eras; rollouts differed by country.
Voice → messaging → internet → broadband → connected sy
Read the rate labels, not just the numbers
Early services, later enhancements and theoretical peaks are distinguished above. These are not equivalent everyday speed measurements.
Actual speeds vary
Network, spectrum, device and conditions affect performance. The 5G figure is a theoretical peak downlink under ideal conditions, not a typical user speed.
kbps = kilobits/s · Mbps = megabits/s · Gbps = gigabits/s · ~ = approximately
VoLTE = voice over LTE · IP = Internet Pr


1980s
1990s
2000s
2010s
2020s
2G
Digital communication
Digitally encoded signal
TECHNOLOGY
Digital cellular
GSM / IS-95
BASIC GSM DATA
9.6–14.4
kbps · circuit-switched
Later GPRS / EDGE: up to ~384 kbps.
KEY FEATURES
• Digital voice and SMS
• Basic mobile data
3G
Mobile internet
Internet data in packets
TECHNOLOGY
UMTS / WCDMA
CDMA2000
EARLY DATA RATES
384 kbps–2 Mbps
Early 3G services
Later HSPA+: up to ~42 Mbps.
KEY FEATURES
• Mobile web and email
• Video calling
4G
Mobile broadband
All-IP voice and data
TECHNOLOGY
LTE
LTE-Advanced
LTE PEAK
~100
Mbps · peak rate
Later LTE-Advanced: up to ~1 Gbps.
KEY FEATURES
• All-IP broadband
• HD streaming and VoLTE
5G
Connected systems
Many connected devices
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